

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Read image
image = cv2.imread('Sample.jpg')
image = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# -----
# Original Image
# -----

# Create aliasing by undersampling
small = cv2.resize(image, (image.shape[1]//6, image.shape[0]//6),
                    interpolation=cv2.INTER_NEAREST)

aliased = cv2.resize(small, (image.shape[1], image.shape[0]),
                     interpolation=cv2.INTER_NEAREST)

# -----
# Anti-aliasing
# -----
```

```
blurred = cv2.GaussianBlur(image, (5,5), 1.5)

small2 = cv2.resize(blurred, (image.shape[1]//6, image.shape[0]//6),
                    interpolation=cv2.INTER_AREA)

corrected = cv2.resize(small2, (image.shape[1], image.shape[0]),
                      interpolation=cv2.INTER_LINEAR)

# -----
# Quantization Example
# -----

levels = 16

quantized = np.floor(image/(256/levels))*(256/levels)
quantized = quantized.astype(np.uint8)

# -----
# Display Results
# -----

titles = [
    "Original",
    "Aliased Image",
    "Corrected Image",
    "Quantized Image"
```

```
]

images = [
    image,
    aliased,
    corrected,
    quantized
]

plt.figure(figsize=(18,8))

for i in range(len(images)):
    plt.subplot(1,4,i+1)
    plt.imshow(images[i])
    plt.title(titles[i])
    plt.axis("off")

plt.show()
```

Original



Aliased Image



Corrected Image



Quantized Image



